

Names, Bindings and Scope

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1 Introduction

In an imperative language, a variable is an abstraction of a memory cell

2 Binding

Binding is associate entity with attribute.

2.1 Possible Binding Times

- **Language design time**
- **Implementation time**
- **Compile time** : Bindings a variable with a type
- **Load time** : Bindings a static local variables
- **Run time** : Bindings a nonstatic local variables

3 Names

3.1 Length

If They too short , they cannot connotative.

3.1.1 Language examples

- FORTRAN I : Maximum 6
- COBOL : Maximum 30
- FORTRAN 90 and C 89 : Maximum 31
- C 99 : Maximum 63
- Java, C++, Csharp : No limits

3.2 Case sensitivity

C-Based language is case sensitive but others are not.

4 Special words

4.1 Keywords

word is used in Certain contexts

4.2 Reserved words

Words cannot be used as pre-defined words , FORTRAN have more than 300 reserved words !!!

4.3 Special characters

- PHP: all variable names must begin with Dollar signs
 - Perl: all variable names begin with special characters, which specify the variable's type
- @ are Ruby: variable names that begin with '@@' are instance variables
- examples : if , else , int

5 Attributes

Variables can be characterized as a sextuple of attributes:

- Name
- Value
- Address
- Lifetime
- Scope
- Type

6 Declaration

two kinds of declarations

- **Explicit declaration** : int x = 5;
- **Implicit declaration** : x = 5;

6.1 Type binding

When? - three kinds of type bindings:

1. Static type binding
2. Dynamic type binding
3. Type inference

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